

Chanuka Algama

PhD Researcher | Spatiotemporal ML, Anomaly Detection, and Interpretable AI

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[Google Scholar](#) | [LinkedIn](#) | [GitHub](#) | [Personal Website](#)

Professional Summary

Computer Science PhD researcher at Tulane University developing scalable and interpretable methods for anomaly detection in complex spatiotemporal data. Experienced in multimillion-record datasets, unsupervised learning, deep learning, geospatial modeling, and machine learning for public-policy applications. Contributor to the IARPA HAYSTAC program, with two papers accepted at ACM SIGSPATIAL 2026.

Technical Skills

Programming and Data: Python, MATLAB, NumPy, pandas, GeoPandas, parallel processing, and memory-aware data pipelines.

Machine Learning: PyTorch, scikit-learn, unsupervised anomaly detection, Deep SVDD, interpretable ML, supervised learning, LLM-based reranking, retrieval-augmented generation, and computer vision.

Geospatial and Statistical Methods: Trajectory analysis, spatiotemporal modeling, remote sensing, spatial econometrics, OLS, spatial regression, Bayesian geostatistical modeling, and geographically structured validation.

Research Engineering: Git, Linux, large-scale experimental pipelines, model evaluation, data visualization, and reproducible research.

Education

Tulane University, New Orleans, LA

Aug 2025–Present

PhD in Computer Science | Advisor: Professor Carola Wenk

Research: Anomaly detection, spatiotemporal machine learning, geospatial AI, and interpretable ML.

University of Kelaniya, Sri Lanka

BSc (Hons) in Computer Science, First-Class Honours

GPA: 3.73/4.00 | Minor: Pure Mathematics

Research Experience

Graduate Research Assistant — Tulane University

2025–Present

IARPA HAYSTAC-supported research

Hidden Activity Signal and Trajectory Anomaly Characterization

- Developed LIAD, a lightweight framework for interpretable, individual-level anomaly detection in movement trajectories, combining Temporal Signature Matching for weekly behavioral changes with an Origin-Return Encoder for trip-level deviations.
- Generated agent-specific explanations from changes in travel distance, speed, dwell time, destination choice, visited locations, and route structure.
- Built memory-aware Python pipelines with parallel processing for datasets containing more than 14 million historical and future trajectory records.
- Evaluated rank-based unsupervised scoring, supervised baselines, and Deep SVDD across synthetic and empirical trajectory datasets.
- Developed Agentic-LIAD, an LLM-based evidence-adjudication and reranking extension.

Undergraduate Research — Optical Flow

- Developed a higher-order differential optical-flow method using a second-order Taylor formulation, perturbation-based correction, and Horn–Schunck regularization; evaluated the method on KITTI and Middlebury optical-flow benchmarks.

Professional Experience

LIRNEasia | Data, Algorithms and Policy Team

Researcher 2024–2025

Junior Researcher 2023–2024

Data Science Intern 2023

- Combined mobile call-detail records, census data, and remote-sensing variables in machine-learning and spatial-econometric models to estimate poverty at fine geographic resolution in Sri Lanka.
- Evaluated OLS, spatial regression, and Bayesian geostatistical approaches using geographically structured validation and decision-centered measures of the models’ ability to identify high-poverty regions.
- Developed an open-source geospatial pipeline to classify built-up and non-built-up areas from satellite imagery and operationalize the UN-Habitat city definition for mapping urban boundaries.
- Analyzed smart-meter records and household-survey data for energy and public-policy research, including household electricity consumption, climate, and energy-access applications.

Selected Publications

Chanuka Algama, Erfan Hosseini Sereshgi, and Carola Wenk. “**LIAD: Lightweight Interpretable Mobility Anomaly Detection via Agent-Adaptive Behavioral Deviations.**” 34th ACM International Conference on Advances in Geographic Information Systems (ACM SIGSPATIAL), 2026. **Accepted.**

Chanuka Algama. “**Drift or Deviation? Continual Detection of Anomalies Amid Changing Patterns of Life.**” ACM SIGSPATIAL PhD and Graduate Studies Workshop, Graduate Student Research Competition, 2026. **Accepted.**

Chanuka Algama and Kasun Amarasinghe. “**Robust Optical Flow Computation: A Higher-Order Differential Approach.**” Proceedings of the 21st International Conference on Computer Vision Theory and Applications (VISAPP), Vol. 3, pp. 373–380, 2026. **Published and presented.**

Chanuka Algama, Merl Chandana, Viren Dias, and Kasun Amarasinghe. “**Informing Public Policy with Machine Learning: Mapping Poverty in Sri Lanka Using Mobile Call Detail Records and Remote Sensing Data.**” AAAI 2025 AI for Public Missions Workshop, 2025. **Published and presented.**

Selected Awards and Professional Service

Second Prize, IEEE SA P2834 Student Challenge Valencia, Spain, 2022

Developed ConvAuther, a conversational-AI system for multifactor authentication; selected as an international finalist.

Dean’s List, Faculty of Science, University of Kelaniya 2017–2019

Recognized for academic excellence in the 2017/2018 and 2018/2019 academic years.